MULTIMACSIM.v1.1 — Interactive Wiring & Simulation Testing Upgrade

Diagnosed Issues in v1.0

1. **Pin Coordinate Displacement Bug:** Pin views were nested inside `componentView` (already positioned at `comp.position`), while calculating `absolutePinPosition()`. This caused pins' click hit-boxes to be placed at double their offset, making them impossible to click and connect.
2. **Gesture Collision:** The canvas background `DragGesture(minimumDistance: 0)` was intercepting mouse-up events and immediately calling `cancelWiring()`.
3. **Tiny Hit Target:** 8px pin hit targets were too small for comfortable trackpad/mouse clicks.
4. **Lack of Simulation Diagnostics / Testing:** Students had no quick way to inspect node voltages on the canvas, check circuit health (floating pins, missing ground), or view a DC operating point summary table.

Solutions Implemented in v1.1 (Completed & Verified)

1. Unified Component Dragging & Pin Wiring Fix

- ✓ **Translation-Based Dragging:** Replaced buggy `value.location` with `startPos + value.translation` snapped to grid, completely eliminating jumping and freezing when moving elements.
- ✓ **Active View Redraw:** Implemented `updateComponentPosition()` with explicit `objectWillChange.send()`, guaranteeing butter-smooth 60/120 FPS component movement with all connected wires dynamically stretching in real time.
- ✓ **Focused Pin Hitboxes (16px):** Scaled pin hitboxes to 16px right at the terminal lead tips so they never obstruct the 60x60 component body.
- ✓ **Effortless Wiring:** Clicking a terminal dot instantly initiates or completes a wire with magnetic snap and green glowing target rings.

- ✓
  Canvas Background Panning: Dragging the empty canvas now smoothly pans the entire schematic (`panOffset`), while clicking deselects.

2. Comprehensive Simulation Testing & Diagnostics

- ✓
  Live On-Canvas Node Voltage Badges: Overlays on every pin displaying instantaneous voltage (e.g. `[ 12.0V ]` , `[ 6.0V ]` , `[ 0V ]`).
- ✓
  Circuit Diagnostic & Health Checker: Toolbar button **"Run Test"** checks for missing ground, floating pins, and closed-loop current flow.
- ✓
  DC Operating Point (.OP) Testing Table: Complete tabular breakdown of all Node Voltages, Branch Currents, and Power Dissipations.